MILITARY SPECIFICATION

FAN, VANEAXIAL, 300 CFM, 400 CYCLE, 416 VOLTS, 0.7 H.P.

1. SCOPE

1.1 This specification covers one type of vaneaxial fan.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-M-7969 - Motors, Alternating Current, 400 Cycle, 115/200 Volt System, Aircraft, General Specification for.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage

DRAWINGS

US ARMY MUNITIONS COMMAND

C5-19-1254 - Fan, Vaneaxial.
D136-14-302 - Tester, Air Flow, Filter Unit, Q31 Assembly

FSC 4240

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies of this publication should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

3. REQUIREMENTS

3.1 Manufacture and assembly. The vaneaxial fan shall be manufactured and assembled as specified on Drawing C5-19-1254.

3.2 Motor. The motor of the vaneaxial fan shall conform to the general requirements of MIL-M-7969 and the following detailed requirements:

(a) Electrical characteristics: 416 volts, 400 cycle alternating current.
(b) Output: 0.7 horsepower.
(c) Speed: 11,275 rpm.
(d) Duty cycle: continuous.
(e) Phases: three.
(f) Minimum efficiency: 70 percent.
(g) Minimum power factor: 0.7.
(h) Inclosure: totally inclosed.
(i) Operating temperature: -65° to 120°F.
(j) Altitude: to be operated only on the earth's surface.

3.3 Air output. The vaneaxial fan shall produce a minimum of 300 cubic feet per minute of air at static pressure of no less than 8-1/4 inches of water when tested as specified in 4.3.4.

3.4 Dynamic balance. The supplier shall certify that the vaneaxial fan conforms to the following dynamic balance requirement: the vibration amplitude shall not exceed $\pm$ 0.001 inch when operated at the rated speed.

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3.5 Weight. The weight of the vaneaxial fan shall not exceed 10 pounds.

3.6 Preproduction. Prior to the start of regular production, a preproduction sample shall be produced in accordance with this specification for examination and test (see 4.2).

3.7 Workmanship. The vaneaxial fan shall be free from burred, cracked, bent, or dented metal sections, scratched or chipped paint; and foreign matter (dirt, oil, or viscous materials).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction.

4.2.1 Sample. A preproduction sample of two vaneaxial fans shall be manufactured using the same methods, materials, equipment, and processes as will be used during regular production (see 6.2).

4.2.2 Examination. Each sample fan shall be examined for all requirements of the applicable drawings and this specification.

4.2.3 Tests. Each sample fan shall be tested in accordance with 4.3.4.

4.2.4 Acceptance/rejection criteria. Each fan in the preproduction sample shall comply with the examination and tests specified in 4.2.2 and 4.2.3.

4.3 Regular production.

4.3.1 Lotting. A lot shall consist of the vaneaxial fans produced by one manufacturer, at one plant, from the same materials, under essentially the same manufacturing conditions.

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4.3.2 Sampling for examination and nondestructive tests. Sampling shall be conducted in accordance with MIL-STD-105.

4.3.3 Inspection.

4.3.3.1 Examination and nondestructive tests. Sample fans shall be examined and tested in accordance with the classification of defects and MIL-STD-105.

4.3.3.2 Classification of defects.

(a) Fan, vaneaxial, 300 CFM, 400 cycle, 416 volts, 0.7 H.P. (Dwg C5-19-1254).

Categories	Defects	Acceptance standards
Critical:	None defined	
Major:	AQL 0.10 percent defective	
101	Air output	4.3.4.
	AQL 1.0 percent defective	
102	Overall length incorrect	
103	Diameter of mounting holes incorrect	
104	Location of flange mounting holes incorrect	
105	Length of leads less than minimum required	
106	Overall diameter of flange incorrect	
107	Workmanship (3.7)	
Minor:	AQL 2.5 percent defective	
201	Marking incorrect	
202	Weight incorrect	

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4.3.4 Air output test. The vaneaxial fan air output shall be determined using Edgewood Arsenal Filter Unit Air Flow Tester, Q31 (D136-14-302).

5. PREPARATION FOR DELIVERY

5.1 Level C preservation and packaging. The vaneaxial fan shall be cleaned, dried, preserved, and packaged to afford adequate protection from the supply source to the first receiving activity for immediate use.

5.2 Level C packing. The vaneaxial fan packaged as specified in 5.1 shall be packed to insure carrier acceptance and safe delivery at first destination for immediate use. Containers shall be in accordance with Uniform Freight Classification rules or regulations of other carriers applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order, all interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

5.4 Repair part. When this item is procured for storage and issue as a repair part, preservation, packaging, packing and marking shall be as specified on the repair part packaging data sheet which is identified by the Federal stock number for the repair part.

6. NOTES

6.1 Intended use. This vaneaxial fan is a component of the M18 gas-particulate filter unit for the hawk battery control central.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Preproduction

(1) Time allowed for supplier submission of samples for Government test and evaluation after award of contract.

(2) Name and address of test facility and shipping instructions when testing is performed by the Government.

(3) Time required for the Government to notify the supplier whether or not to proceed with production.

6.3 Supersession data. This specification supersedes Chemical Corps Purchase Description No. 197-54-730, dated 16 June 1960.

Custodian:

Army - MU

Preparing activity:

Army - MU(EA)

Project No. 4240-A220

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